Chapter 8

Let's Give Life to Pictures

Have you noticed the picture of a plant and a puppy in the lower right corner of the right page of our ICT textbook ?

Turn the page. You can see the same picture at the same place on the next page too.

And what about on the next page ?

Turn the pages one by one. Are the pictures same on every page?

Now, quickly turn the pages from the first page to the last page. What do you see?

It looks like as if the plant is growing and the puppy is looking at the plant wagging its tail, right ?

Have you ever wondered how this is possible ?

Notice that each page has different pictures of the puppy and the plant. These include different stages of the plant growing and the puppy wagging its tail.

When the continuous and sequential images of the movement of the puppy and the plant were rapidly captured one after the other by our eyes, they were perceived as a motion.

It is because of the unique feature of our eyes called 'persistence of vision' that the still images in the textbook seemed to come to life. It is by making use of this feature that the technology of animation came into existence.

Persistence of Vision

An image that falls on our eyes remains on the retina for a short time (one sixteenth of a second). This feature is called Persistence of vision. When another image falls on the retina during this time, we experience the second image as a continuation of the first image.

Let's Make an Animation Video

Animation is a field that quality interests all of us. Earlier we were familiar with animation through cartoon movies on television channels, but today we constantly see and make use of it in educational videos, advertisements, reels, and greeting messages.

In the past, animation films were made using painted plastic sheets. With the advent of computer technology, the animation technology has become more popular and the production has become relatively easier.

Let's Make an Animation Video

Phenakistiscope

Phenakistiscope

Phenakistiscope is the first device used to simulate animation. Sequential images of the various stages of an object's movement is drawn on the surface of a circular hardboard. When it is rapidly rotated using a stand, the object on the hardboard appears to move. Based on this technique, animations were created in the beginning of 19th century. The possibilities of animation were introduced with the help of devices like phenakistiscope during 1830s.

Didn't you notice a small animation hidden within the ICT textbook ? This is a Flipbook animation. Instead of that, shall we try to create a real animated movie ourselves ?

To create an animation, some preparations are necessary. Just like in the making of a film, play, or a documentary, an animated movie is also created through various stages.

Flipbook

A flip book is a type of illustrated book. It is also known as Flickr book and Kineograph. They are made by adding a series of images that change very gradually from one page to the next. When we look at these pages continuously, we feel that the images are moving.

There are websites and software packages available on the Internet today that can convert PDF pages and digital videos into desired Flipbooks. Search them in the Internet and try out.

You have learned in previous classes that for preparing a film or documentary, it requires a story, a script, characters, and dialogues.

Find out what all preparations are needed to prepare an animated film and complete the list given below.

- A theme is required
- Story based on that theme should be prepared
- The order in which the events in the story are presented, or the scenes should be determined.
- A story board including the details of each scene should be prepared
- .....
- .....
- .....

We have understood what all preparations are necessary to make an animation film. Based on this, let us now learn how an animation film is made.

An animation film should be made based on a particular theme or concept.

Now closely observe the flipbook animation included in the textbook. What is the theme of this animation? Discuss with your friends and find out.

Shall we consider this animation as a specific scene from an animated film that promotes environmental protection?

If so, think about what other situations there might be.

The plot of an animated film based on this theme, namely 'Luna', as prepared by some friends, is given below.

Luna (Animation Film) – the Storyline

A plant that has withered from not having enough water. A small puddle in the distance.

Small clouds in the sky.

A puppy (Luna) comes near the puddle.

The puppy makes a channel in the soil with its front paws and

brings the water to the plant.

You have seen the staryline of the animated film.

Which characters should be included when this story of Luna is made into an animated film?

- Luna
- .....
- .....
- .....

You have seen that a large number of images are required to create an animation. The story should be divided into different scenes based on the background and context in which they take place. The images should be designed with detailed illustration of the characters in each such scene.

Designing characters is a task that requires utmost care. It requires skill and expertise in drawing.

Character Design

Character design is a very important step in animation production. It is the process of creating the characters in a way that suits the needs of the story. While designing characters, it would be good to take the following aspects into consideration as well.

- Variation in the shape of the characters
- Appropriate selection of colours
- Drawings that can be easily modified

With the advent of software-based animation production, creating drawings for animation has become much easier. Animation software usually includes built-in tools for drawing. It is also possible to create drawings using dedicated drawing software and then importing them into animation software to create the animation.

Different Types of Animation

Animations are created in 2D and 3D formats. 2D animation is a method of creating animation based on 2D images. Walt Disney's The Jungle Book, Bambi, Cinderella, etc. are famous 2D animated films. Open Toonz, Tupi Tube Desk, Synfig Studio, Adobe Animate, etc. are software that helps in creating 2D animations.

3D animations are created by preparing 3D shapes of the characters and backgrounds. Movies like Big Buck Bunny, Shrek, and Wall-E were created in this way. 3D animation is also widely used in games today. Blender, Autodesk Maya, and Adobe Dimension are examples of 3D animation software.

When Luna's story is divided into individual scenes, which are the characters to be included in the first scene? Discuss.

The next step in creating the animation is to convert the prepared story into a storyboard. A storyboard is detailed preparation of each scene in advance..

Based on the first scene of Luna's story, a sample storyboard has been provided below (Table 8.1). Discuss with your friends and prepare the storyboards for the other scenes using this sample as a model.

Titile : Luna				**Page Number : 1**		
Scenes	Background/ Characters	Action	Dialogue	Music	Time	Remarks
Scne 1	Withered plant, small puddle, sky	Clouds are moving in the sky		Background music	5 second	
Scne 2						
Scne 3						

Table 8.1 Model of a Storyboard

You've seen the storyboard for the first scene. Now let's create the animation based on this.

Towards Motion

We shall first prepare the scene that includes the movement of the clouds as described in the first scene. Let's create this animation using the OpenToonz animation software available on the school computer.

In our scene, we need a withered plant and a small puddle in the background. This animation will be possible only if we have images of all these.

We have already learned in previous classes how to create images using Krita, GIMP, and Inkscape. We can prepare the images using those software tools, or we can draw them directly in OpenToonz to create this animation.

OpenToonz

OpenToonz is a free animation software based on "Toonz" software developed by Digital Video S.p.A. in Italy. It provides tools for drawing, animation, and sound.

The Japanese animation film 'Belle' released in 2021 was created using OpenToonz software.

Open OpenToonz on your computer and create a project for the first scene.

To Create a Project in OpenToonz

- Open OpenToonz software
- In the OpenToonz Startup window that opens, click New Project to create a new animation project.
- Enter a name for the project (here Luna) and click OK (figure 8.1)
- Click Create a New Scene in the OpenToonz Startup window
- Then, enter the name of the scene and video settings in the window that appears. (Figure 8.2)

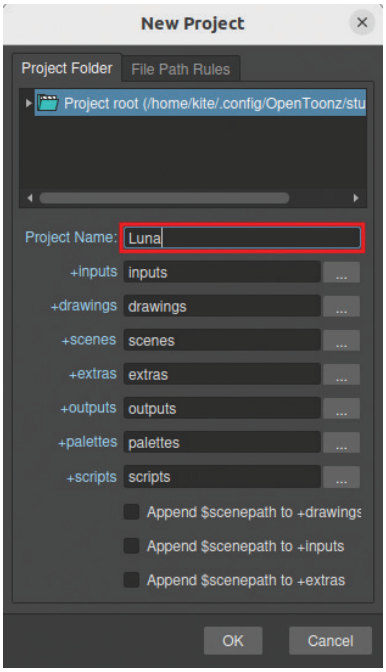


Fig. 8.1 New Project Window

The animation we create will be available to us in the form of a video file. When creating an animation video, the quality and frame rate of the video files need to be set in advance in the video profile.

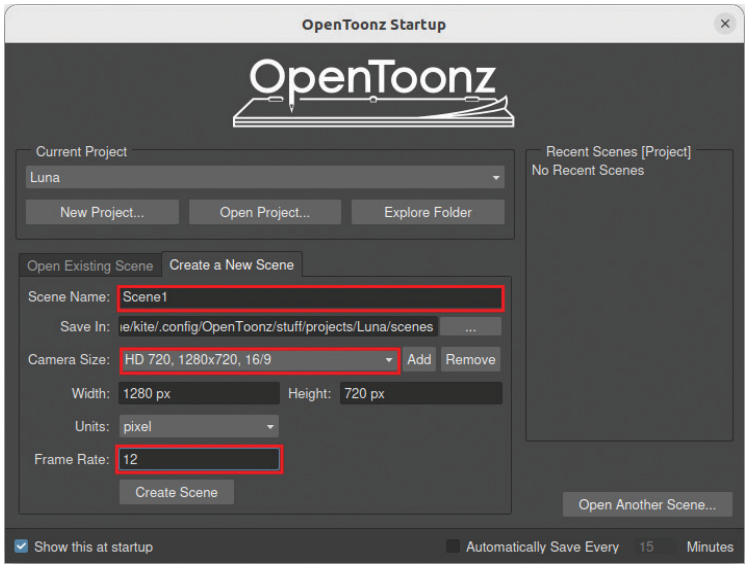


Fig. 8.2 OpenTooz Startup Window

The necessary settings for this also need to be entered in the OpenToonz Startup window.

A model of the settings that can be entered for video profiles in OpenToonz is given below.

Scene Name : Scene1
Camera Size : HD720,1280x720, 16/9
Units : pixel
Frame Rate : 12

Similarly, you may configure necessary settings for your animation project.

Once you provide the necessary settings for the first scene and click on Create Scene, the window required to create the animation will appear.

Did you get the OpenToonz window ? (See Figure 8.3)

Use the mouse scroll wheel to adjust the view so that the entire canvas is visible.

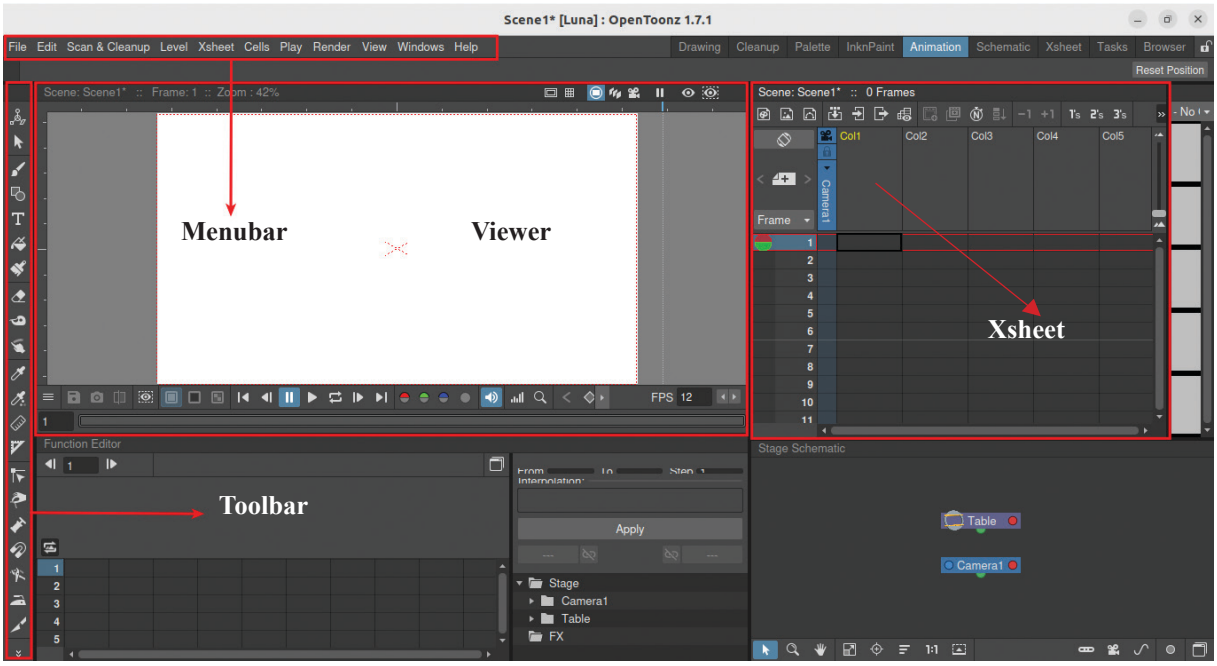


Fig. 8.3 OpenToonz Startup Window - Animation Room

Frame Rate

In animation, a frame can be considered as a still image. Animation is possible by displaying many such frames continuously. The number of frames displayed in one second is called Frame rate (Frame per second, FPS). This is a factor that determines the naturalness of the movement in the animation. We can set FPS according to the needs of each animation.

Let's Set the Background

According to our storyboard, in the first scene, we need to create a withered plant, a puddle, and clouds moving across the sky.

As mentioned earlier, animations can be created in OpenToonz by drawing the necessary images directly or by including pre-made images in the scene. Although simple images can be drawn using the drawing options provided in OpenToonz, it is more convenient to prepare and use complex images with the help of other drawing software that we are familiar with.

Xsheet

Xsheets are a feature available in OpenToonz for organizing the content of animation scenes. It is a concept similar to timelines in video editing software. Xsheets are arranged in different columns, with each column functioning as a separate track. Frames are arranged in the columns as cells.

Some of the images, image sequences, and audio files for preparing our animation are provided in the folder namely Class_10/Animation in School_Resources on the computer. These files may be used accordingly while creating the animation..

Arrange the Resources

In order to include the resources in OpenToonz, copy the Animation folder provided in School_Resources/Class_10 and paste it into the Documents folder in Home.

To begin with we will create background for the first scene. Check the image namely bg.png provided in the Animation/Scene_1 folder. Let's set this image as the background for our scene.

This should be done by opening the Animation Room in OpenToonz software.

Rooms in the OpenToonz

OpenToonz has separate rooms for different activities. These are also known as workspaces. When you open **OpenToonz** for the first time, the **Drawing** room opens.

Animation: This room is where you animate the position, rotation and size of the current objects.

Drawing: This room is used to draw, colour, and edit the drawings.

Open other rooms to find out what they are used for.

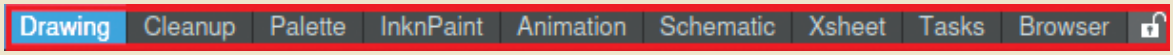


Fig. 8.4 Various Rooms in OpenToonz

To Include Image in OpenToonz

The background image should be included in the first frame of the first column.

For this,

- Select **Animation** room from the **OpenToons** window.
- Click on the first frame of the first column in the Xsheet.
- Click on **Browser** Workspace in the menu bar.
- Then browse to the folder where the images are stored from the window that appears.
- Select the image **bg.png,** Right-click and click **Load.**
- Click **Load** in the dialogue box that appears.

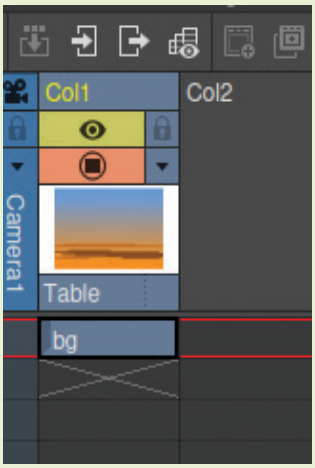


Fig. 8.5 When Imag
Included in OpenToonz

The background image has now been set in the window. Now, let's return to the Animation workspace.

Currently, the image we inserted appears only in the first frame. For the animation to work properly, this image must be included in the other frames as well.

We are creating a 5 second long animation with a frame rate of 12 frames per second as our first scene. So, think about how many frames are to be included in the first scene.

This image in the first frame has to be included in the remaining frames too (60 frames in total), right?

OpenToonz has a feature called Click and drag to repeat selected cells to copy the content in one frame to other frames. Try using this to include the background image in other frames.

Once the work is completed, don't forget to click File → Save All to save your work.

How many Frames?

For an animation of 1 second in 12 FPS : 1 x 12 = 12 Frames

For an animation of 5 second in 12 FPS : .....

To Include Images in Other Frames

- Click on the first frame in Col1.
- Click on the small button below this spell - Click and drag to repeat selected cells.
- Drag the button to include the image up to the required frame.
- This can also be done using the Repeat option too.

Now the background image has been added to all the frames of the first scene.

According to our storyboard, clouds are moving in the sky in the first scene.

Let's see how can this be done.

Draw a Cloud

The background is prepared using an image drawn in another software. What about drawing the cloud in **OpenToonz** itself ?

Simple images can be drawn with the help of the drawing tools of the OpenToonz software.

The background is in the first column. It is possible to give separate animations to the objects, if they are drawn in separate columns.

So, let's draw the cloud in the second column.

Repeat Button

The **Repeat** button can also be used to include the contents of a cell in other frames. To do this, select the image and click the **Repeat** button. In the window that appears, enter the number of frames and just click the **Repeat** button.

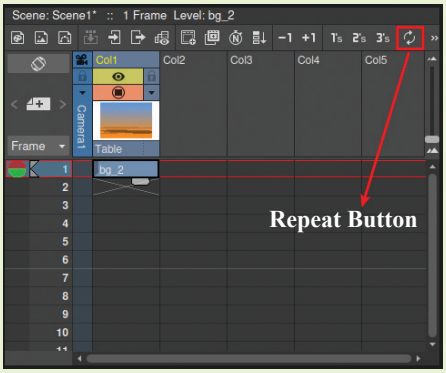


Fig. 8.6 Repeat option in OpenToonz

OpenToonz allows you to create images as raster files or vector files. Here we are drawing the cloud as a vector image.

To draw vector images, you need to set up the Vector Level window. Create a new vector level in the second column using the following steps.

To Create Vector level

- Select the Animation Room.
- Click on the first frame in the column (here - Col2) you want to include from the Xsheet.
- Then, right-click here and select New Level.
- In the New Level window that appears, enter a name in the Name field (Cloud).
- Click the drop-down menu in the Type field and select Toonz Vector Level.
- Click OK.
- With this, you will see that a vector level named Cloud has been created in the selected frame.

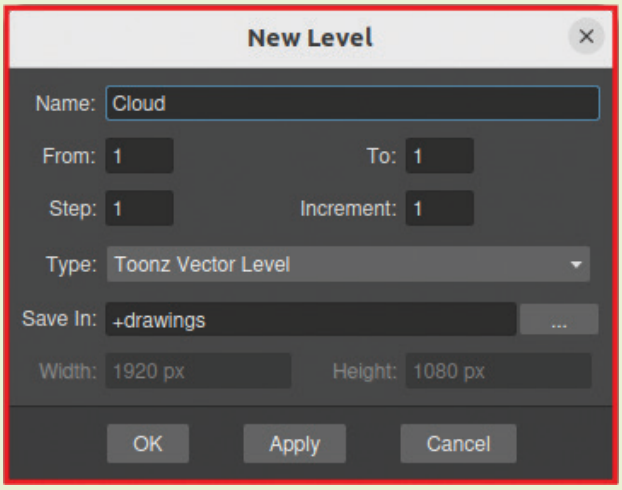


Fig. 8.7 OpenToonz New Level Window

A vector layer is created to draw the cloud. Now, draw the cloud using the drawing tool.

We are now in the Animation room. Find out the room for drawing the images.

To draw images in OpenToonz

- Select Drawing Room.
- Select Brush tool from the Toolbox to draw images.
- Then draw the cloud.
- Select the required colour from the Style Editor (Figure 8.9).
- The drawn image can be erased using Eraser Tool.
- Click inside the shape using Fill Tool to add colour.
- To get a new colour, click New Style and select the desired colour.
- Adjust the drawn image to the required size using the Selection Tool.

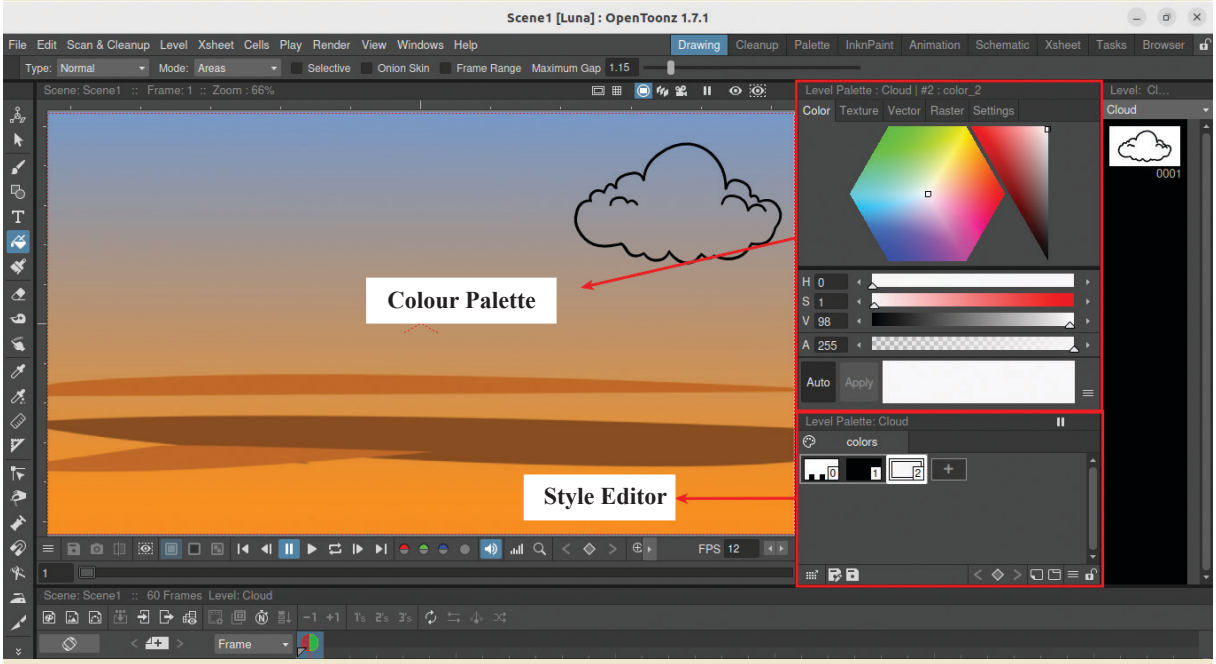


Fig. 8.8 OpenToonz Drawing Rooml Window

Try varying the Size, Accuracy, Smooth etc., while selecting the Brush tool to draw the image. What difference does it make?

Have you finished drawing the cloud ?

Now, we need to include this image in all 60 frames of the scene. Try doing it the same way as in the previous activity.

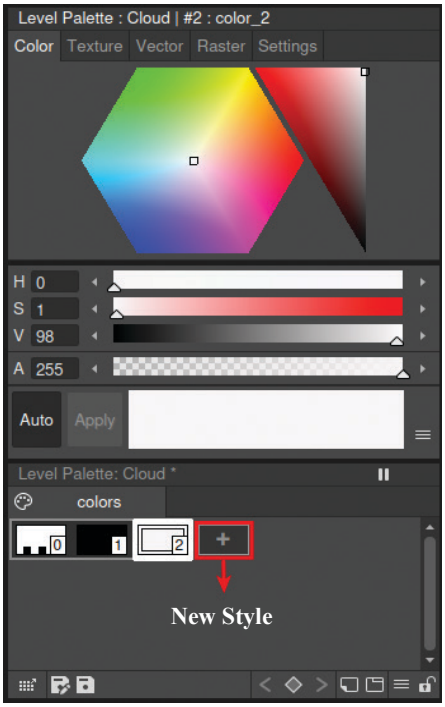


Fig. 8.9 Style Editor Window in OpenToonz

We have already seen some of the tools in OpenToons. What are the other tools in the toolbox? Try them and complete Table 8.2.

Fig. 8.10 OpenToonz Toolbox

Tool	Name	Tool Use
		To adjust the position, rotation, size of the objects
	Selection Tool	
		To draw geometrical figures
	Type Tool	

Table 8.2 Some of the Tools in OpenToonz and their Uses

Let's Add Motion to the Images

We need to create an animation in which a cloud moves from one side of the sky to the other. Let's see how to create this animation.

The movement of the cloud from one side to the other should be completed in 60 frames. In order to make this possible, the position of the cloud must change frame by frame according to its path of motion. That means we need to draw 60 different images, that differ in the position of the cloud drawn on the canvas, from one side to the other.

Instead, if we determine the position of the cloud in the first frame where the movement begins and the position in the frame where the movement ends, what if there was a technique where the software itself prepares the position of the cloud in each frame between them? OpenToonz also has the ability to create animations using a technique called tweening, which is commonly available in animation software. Try creating an animation of a cloud moving from one side to the other using Tweening animation.

Key Frames

When using tweening to create animation, the software must be able to predict the position and size of the object in each frame. This is based on the objects in some key frames that the animator determines. Keyframes are frames that are determined as important positions in the direction of movement in the animation. In OpenToonz, tweening animation can be created based on the Position, Scale, Rotation, Shear, and Center of the objects.

To Prepare Animation in OpenToonz

- Click on the first frame in the column (Col2) where the cloud is drawn.
- Select Animate tool from the Toolbox to give animation.

The software should set the position of the clouds in the other frames based on the position of the cloud in the first and last frames (keyframes). This is done using a technique called Position Tweening.

- Select Position from the animation option.
- In the first frame, adjust the position of the cloud on the canvas to where the movement begins.
- Now, the first frame has been set as a keyframe.
- Next, click on the sixtieth frame and adjust the position of the cloud on the canvas to where the movement ends.
- With this, the sixtieth frame has also become a keyframe. Our object is now ready for animation.

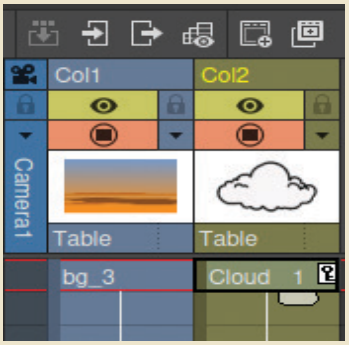
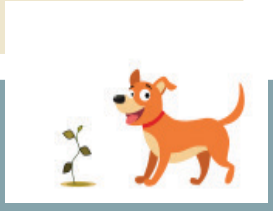


Fig. 8.11 when keyframe is set



The cloud animation has been created with the help of position tweening.

Press the Play button below the canvas. Does the animation work?

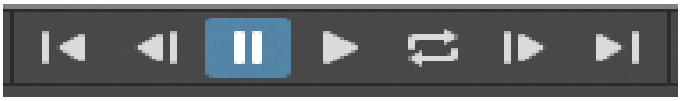


Fig. 8.12 Video Controls

Now, draw another cloud in a new column and give movement. Make the size of the cloud slightly different from the one you drew earlier...

We made the animation possible here by setting keyframes. Can we identify the frames that have been converted into keyframes from the columns in the Xsheet ?

Check the frames and find them.

Save the work using File - Save All. Files in OpenToonz are saved as project files with .tnz extension.

Let’s Include Background Music

In movies, there are different ways to communicate effectively with the audience. The most important of these is the use of sound and music. Shall we include some of these elements too in our small animation movie.

A file named bg_music_1.mp3 is provided in the Animation/Scene1 folder on your computer for background music in the first scene. Add this file to a new column (Col4) and check the animation of the first scene.

We need to use the same method to add sound as we did for the image in the first column. However, since the sound file included in the first scene will also appear in other frames according to its duration, we only need to repeat them if necessary.

Let's Make a Video

A small step of including background music in our first scene has been completed.

Now it is saved as a project file of the OpenToonz software on the computer. We can see the animation only if we open it in the software used to create it.

If we export (render) the prepared animation as a video file, we can run it on the computer and mobile phone without the help of the OpenToonz software. We have been familiar with this method when we made projects in video-audio editing software also.

Use the Render option in OpenToonz to prepare our animation in mp4 video and save it in your folder.

To Render Video in OpenToonz

- From the Menu Bar ൧ Render → Fast Render to MP4.
- The video will be saved as MP4 after rendering.

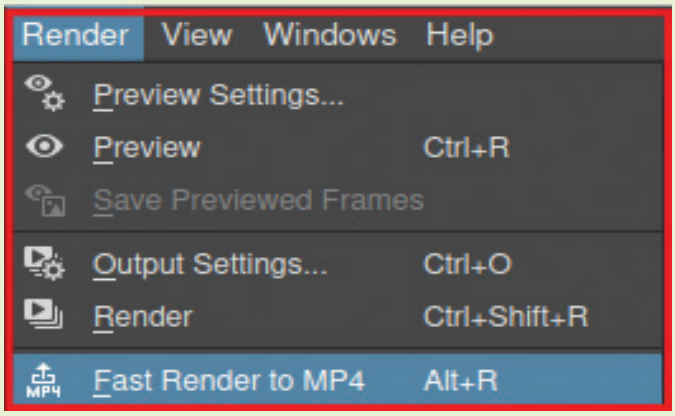


Fig. 8.13 Render Window

Animation Using Image Sequences

We have completed all the tasks of the first scene according to the storyboard. With this, we have become familiar with creating animation using images.

OpenToonz has the facility to create animation using image sequences instead of images also. For this, we need to prepare image sequences in advance. Preparing and using image sequences that contain the characters to enable subtle expressions and movements will make the animation work in OpenToonz easier. Even if the scenes are to be reused, their image sequences are more convenient.

Have you completed the storyboards for the other scenes of the animation you are going to prepare?

Below is the second scene of the storyboard prepared by Daya and her friends based on Luna's story. Let's prepare it with the help of the image sequences..

Scene	Background / Characters	Action	Dialogue	Music	Time	Remarks
Scene 2	Water, plant, Luna	Luna brings water from the puddle close to the plant		Background music	5 second	

Table 8.3 Luna –Storyboard Model (Scene 2)

See the Storyboard. We have to show Luna bringing water from the puddle to the plant. What images are needed to prepare this ?

For this, background image including the plant, image of puddle and the image sequence of the motion of Luna are given in Animation/Scene_2 folder.

While Including Image Sequence

When image sequences are added, we can see that the images have been inserted into as many frames as there are images in the sequence. These images can be inserted into other frames using the Repeat option.

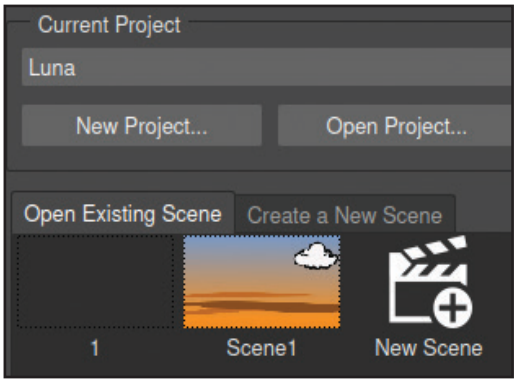


Fig. 8.14 Current Project

We shall prepare the second scene using the images from Scene_2 folder and image sequence from Scene_2/ Sequence folder.

The second scene can can prepared in the same way as we have completed the first scene.

If the OpenToonz have been closed, open it again and ensure that the current project is Luna in the Startup window (Fig 8.14)

- Click on the New Scene icon to begin the new scene.
- Give the name as Scene2
- Give the Camera Size, Units, Frame Rate same as the first scene.

Now, complete the second scene as per the storyboard.

- Include the background image in the first column and the image of water in the second column.
- Include the image of background and water into all the frames using (Repeat).
- Include the image sequence to the first frame in the third column.

Image sequence is to be added in the same way as image is added. But here, as many frames as the number of images in the sequence will be formed instead of one frame.

While Preparing Image Sequences

The file names of the images should be given as in a pattern while preparing image sequences to include in OpenToonz. Let's check the image sequence provided in the Sequence folder. When including such image sequences in the scene, only the common name in the image file name will be displayed in the Browser Room of OpenToonz.

For example, when including an image sequence with names luna _01.png, luna_02.png, luna_03.png, ... only its common name, luna_.png will be displayed in the Load level window. The number of image frames included in the image sequence can be seen in the Frames section.

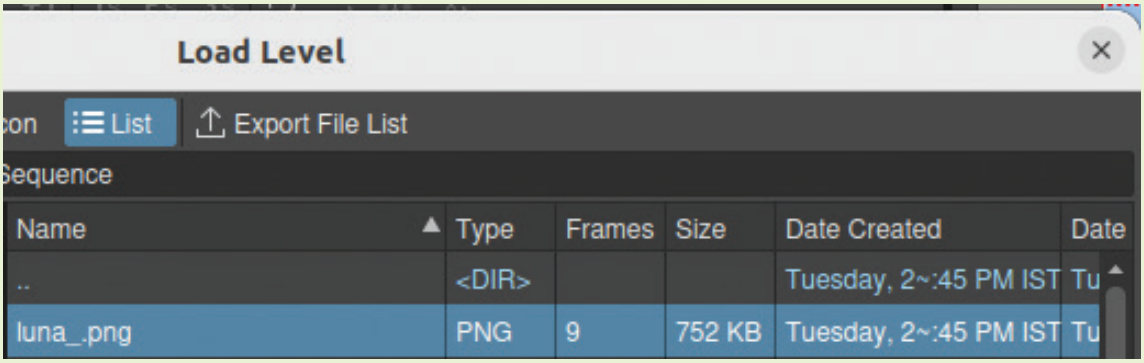


Fig. 8.15 Load Level Window

So, image sequence has been added to the first frame of the third column.

The scene in the sequence shows stagnant water being carried by Luna through the channel towards the plant. This scene will be complete only if this sequence is included in the remaining frames as well.

Use the note below to include the sequence of images in all frames.

To Insert the Image Sequence into Frames

- Click on the frame containing the first image in the sequence.
- Press and hold the Shift button and click on the frame containing the last image in the sequence
- Then, use Click and drag to repeat selected cells to insert the image into the entire frame
- Play the animation using the Play Button.

Animation Opportunities

Animation is widely used today in various fields such as film, advertising, industry, education, healthcare, gaming, etc. It is desirable for animation designers to have expertise in more than one field along with technical expertise. Animation is also a great career field today.

Have you tried running the animation with the image sequence included?

Is the animation of Luna directing the stagnant water, in the correct form now?

Use the Animate tool to adjust the size and position of the image of water in the second column and the position and size of Luna in the image sequence and then play the animation.

You can make the scene more attractive by adding background sound to the scene and an animation of a cloud moving in another column.

Once all the activities are completed, don't forget to render and save the animation video in your folder.

Let's Combine Scenes

The next step is to combine the animation scenes we have prepared into a small animated movie. For this, you can use any video editing software available on Kite GNU Linux.

You may use the Kdenlive software, which you have learned in the previous class. Don't forget to add title slides while preparing the movie.

If you need to make any changes to the prepared scenes, just open the previously saved project file (.tnz), make the necessary changes, and then render the video.

Let us Assess

- The names of some rooms (workspaces) in OpenToonz software are given. Note down their uses.

Name of the Room	Use
Animation	
Drawing	
Browser	

- Given below are the icons of some important tools in OpenToonz. Complete the table with their uses.

Tool	Use

Extended Activity

1. Create the third scene of the animated film Luna using OpenToonz. Combine this with the first and second scenes you created earlier and make a complete film. (Resources for creating the animation are in the folder Animation/Scene_3).
2. Create an animated film to be shown on World Anti-Drug Day based on the slogan "Say No to Drugs". Create the necessary images in Inkscape or GIMP.